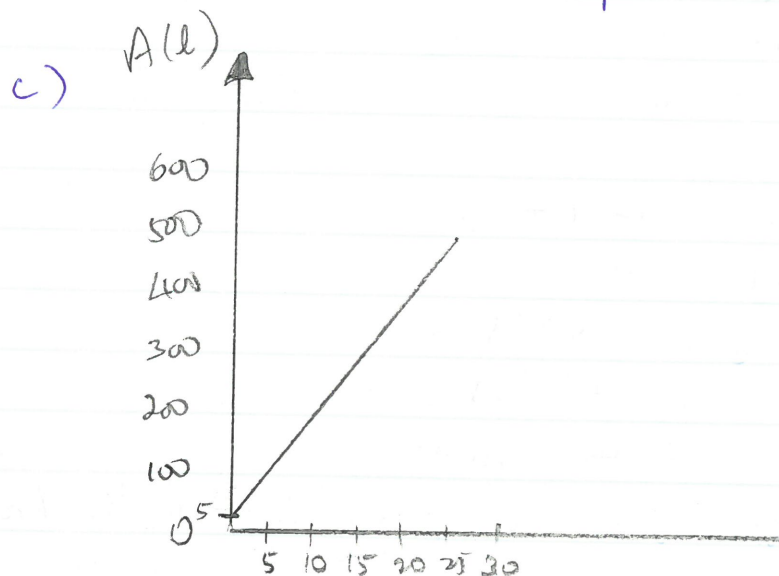


3. $A = 20t + 5$

A = amount of water in litres
 t = time in minutes

a) Both variables have the power of 1.
 The equation is in the form of $y = mx + c$
 which is a linear equation.

b) y-intercept is 5
 This is the initial amount of water, 5 l,
 which is in the pool.



d) $A = 500$
 $500 = 20t + 5$
 $20t = 495$
 $t = \frac{495}{20}$
 $= 24.75$
 $\approx 25 \text{ min}$

It will take 25 minutes to fill the pool.

4. a) $m = 40 \text{ cm}^3/\text{s}$

$$c = 100 \text{ cm}^3$$

$$A = 40t + 100$$

b) y-intercept = 100

This is the initial amount of air (100 cm^3) in the ball.

c) t is in second

$$2 \text{ min} = 2 \times 60 = 120 \text{ s}$$

$$A = 40(120) + 100$$

$$= 4800 + 100$$

$$= 4900 \text{ cm}^3$$

d) $100000 = 40t + 100$

$$40t = 99900$$

$$t = 2497.5 \text{ s}$$

$$\approx 41.625 \text{ min}$$

$$\approx 42 \text{ min}$$

It will take 42 minutes to fill the ball.